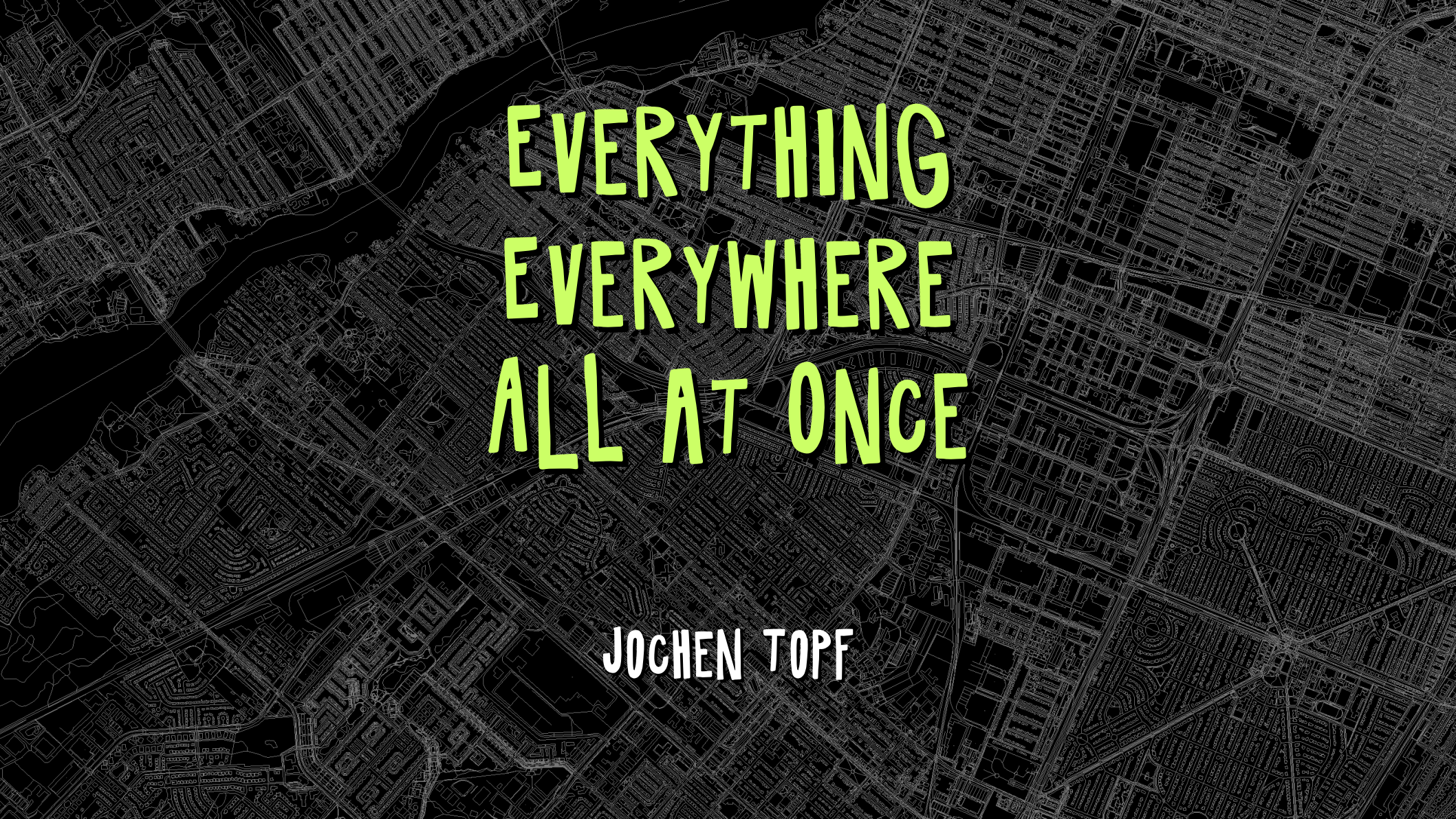


An aerial photograph of a city, likely Berlin, showing a dense grid of streets and buildings. A river, the Spree, flows through the upper left portion of the image. The overall tone is dark, with the city features appearing in shades of grey and black.

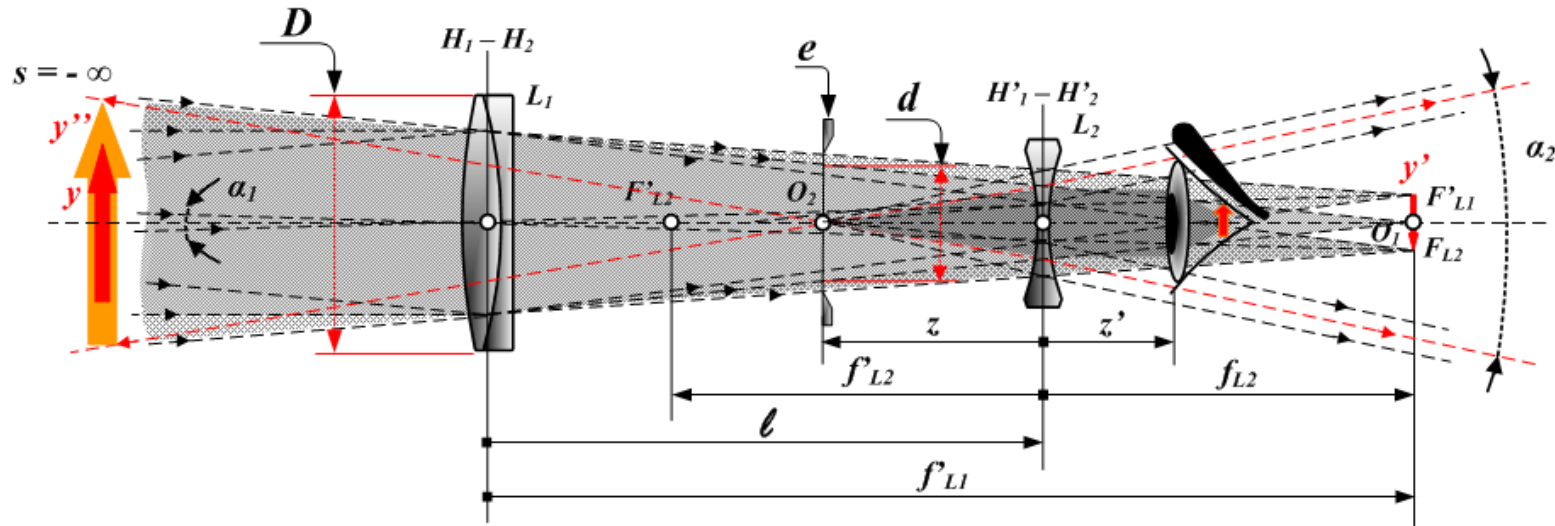
EVERYTHING EVERYWHERE ALL AT ONCE

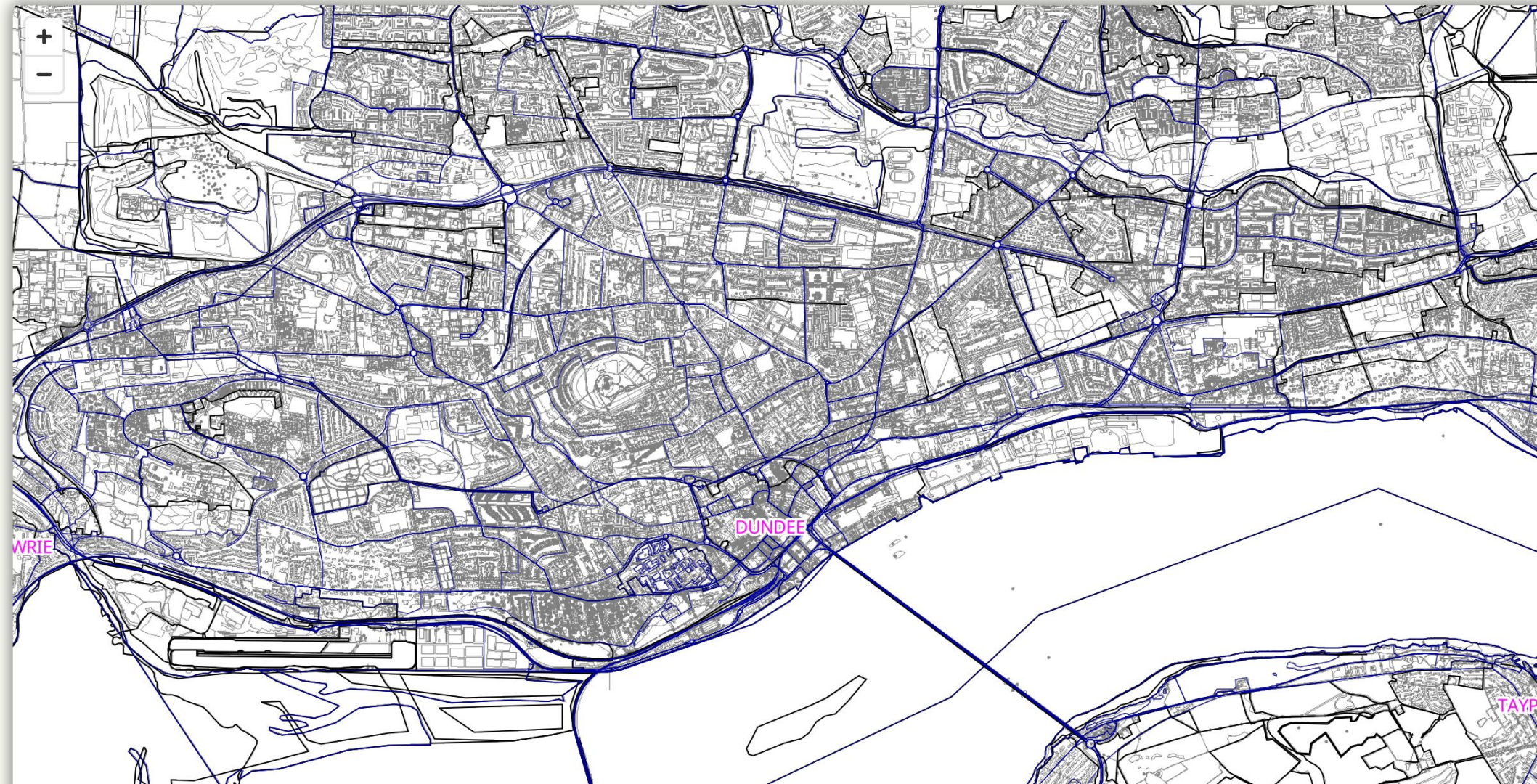
JOCHEN TOPF

we have got this
amazing
collection of geodata...

how can we see it all?

With a Spyglass!





how this came about...

vector tiles
with **all** the data!

how hard can it be?

Prior Art

Mapbox: OSM QA tiles (10 years ago!)

<https://osmlab.github.io/osm-qa-tiles/>

Matt Whilden: Vector Tiles for WA

<https://community.openstreetmap.org/t/demo-full-data-vector-tile-map-with-basic-filtering-for-qa/111351>

okay, it seems we can do this...

but how useful is it?

tiles with everything are
too large and
too slow to process
for common maps

Goal

not just tiles,
but a
complete tool
for mappers

Goals

planetwide
minutely updated
show every tagged object
easy to explore data
reasonably fast to use

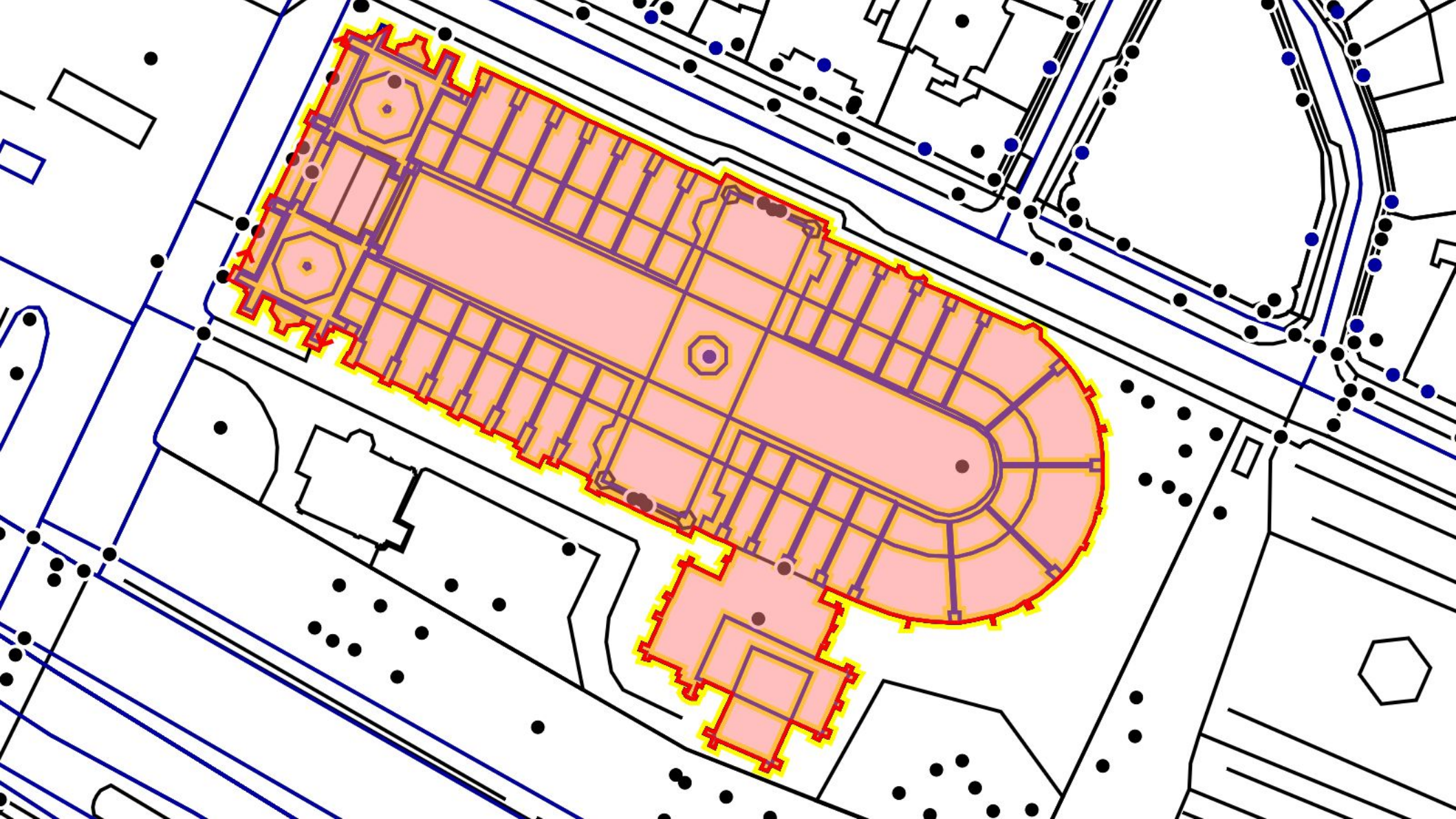
Non-Goals

not a basis for just any map

not for high loads

not for small screens

not showing anything outside Web Mercator area





Selected: 20 (click to lock selection)

2
route 2446626

colour	#007a59
duration	02:06
fee	yes
from	Sacomã
name	477A-10 Term. Pinheiros
network	SPTrans
network:	Q7392689
wikidata	
network:	pt:São Paulo Transporte
wikipedia	
opening_hours	Mo-Su 04:00-25:00
operator	Via Sudeste Transportes S.A.
public_transport:	2
version	
ref	477A-10
route	bus
to	Terminal Pinheiros
type	route
website	https://sptrans.com.br/itinerarios/linha/?cdp=175786

3
route 3200573

colour	#007a59
duration	02:04
fee	yes
from	Ipiranga
name	874T-10 Lapa
network	SPTrans
network:	Q7392689
wikidata	
network:	pt:São Paulo Transporte
wikipedia	
opening_hours	Mo-Su 04:00-24:40
operator	Via Sudeste Transportes S. A.
public_transport:	2
version	
ref	874T-10
route	bus
source	http://www.sptrans.com.br

Background

(None) ▾

50% ▾

Overlays

✓ B ✓ P

Object types

✓ N ✓ W ✓ R

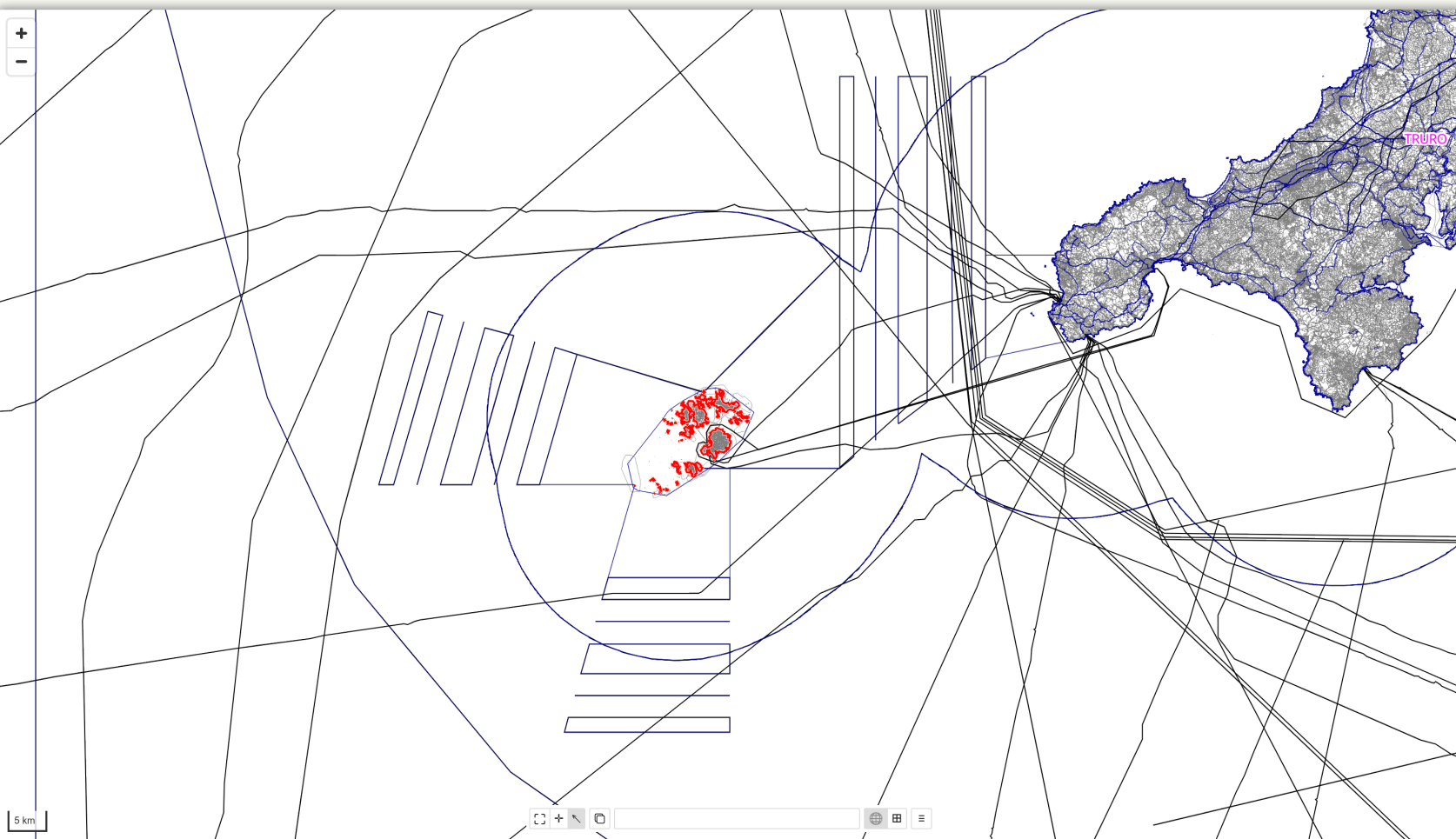
Tag filter

Key:

Value:

Filter

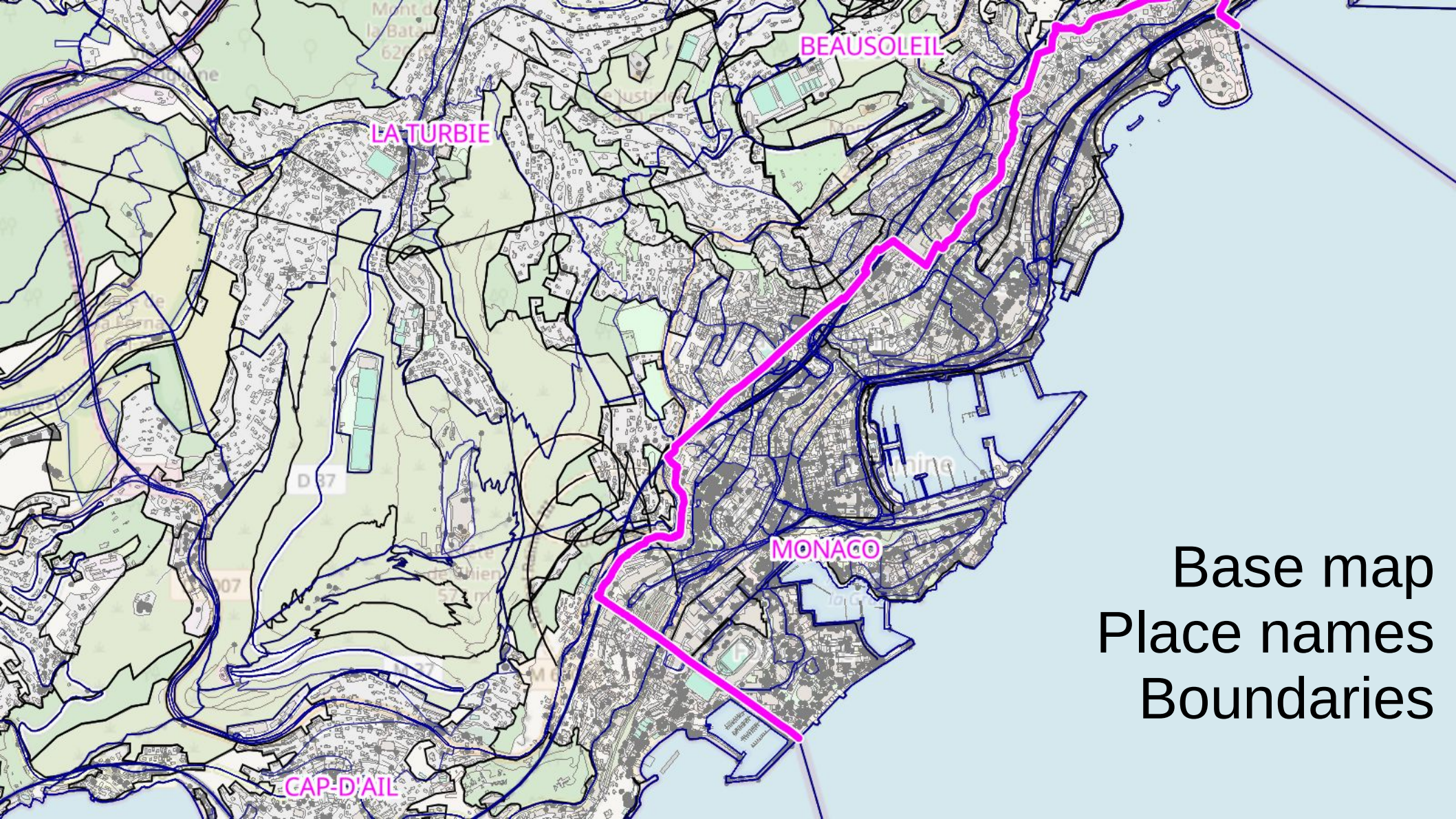
Clear



Selected: 1 (click to lock selection)

1 boundary 158039

admin_level	6
alt_name	Scilly Isles
alt_name:cy	Ynysoedd Syllan
boundary	administrative
council_name	Council of the Isles of Scilly
designation	unitary_authority
ISO3166-2	GB-IO
name	Isles of Scilly
name:af	Scilly-eilande
name:ang	Sylllega
name:ar	جزر سيلى
name:arz	رر سيلى
name:ast	Islles S
name:be	Cini
name:bg	Острове
name:br	Inizi Sill
name:ca	Illes Scilly
name:ceb	Isles of Scilly
name:cs	Scilly
name:cy	Ynysoedd Scilly
name:da	Scillyøerne
name:de	Scilly-Inseln
name:el	Νησιά Σιλί
name:en	Isles of Scilly
name:eo	Insuloj Scilly
name:es	Islas Sorlingas
name:et	Scilly saared
name:eu	Scilly uharteak
name:fa	مجمع الجزایر سیلی
name:fi	Scillynsaaret
name:fr	Îles Sorlingues
name:fy	Scilly-eilannen
name:ga	Na Scillí
name:gl	Illas Scilly
name:he	י"ד סילי
name:hi	सिली द्वीपसमूह
name:hr	Scilly
name:hu	Scilly-szigetek



Base map
Place names
Boundaries

Architecture

OSM planet file/change files

→ osm2pgsql

→ PostgreSQL/PostGIS database

→ Web Server

→ Web Application

Database

all tagged nodes, all ways, relations (members)

all tags

point/linestring/polygon geometries

(can be extended to also contain attributes)

osm2pgsql

config file using Lua code

Themepark!

<https://osm2pgsql.org/themepark/>

re-use boundaries from Shortbread

generalization/post-processing support

High Zoom Levels are Easy

just put everything in the tiles

Mid Zoom Levels are Okayish

remove “very small stuff”

possibly simplify geometry

Low Zoom Levels are Difficult

can't just remove data

generalization generally not possible

tippecanoe? (can't do updates)

<https://github.com/felt/tippecanoe>

Low Zoom Levels are Difficult

different data and different density of data
throughout the world

tiles need to fit together (so have to process
everything with the same settings)

What about raster?

we have upper limit on tile size

relatively simple and fast

we can do that in the database!

What about raster?

but not interactive

→ compromise for small zoom levels

Raster support in PostGIS

```
CREATE EXTENSION postgis_raster;
```

create empty rasters per tile:

```
SELECT ST_AddBand(ST_MakeEmptyRaster(...));
```

fill rasters with data:

~~ST_AsRaster()~~ → ST_SetValue()

Raster support in PostGIS

merge rasters from tiles for lower zoom levels
(not ideal, can we do better outside database?)

create PNGs on the fly:
`ST_AsPNG()`

Also need GeoJSON

for single object

includes bbox
includes children

Server

has to turn data in database into

vector tiles

and

raster tiles

and

GeoJSON

Server (1st try)

pg_tileserv (for vector/raster tiles)

pg_featureserv (for geojson)

nginx reverse proxy (for rewrite shenanigans)

Server (2nd try)

how hard can it be to write my own?

written in Go

Go is made for creating web services
only external lib used is for database access

Server (2nd try)

handles everything
in <400 lines of code

+

about 330 lines of SQL

```

{{- $extent := 1024 -}}

WITH
bounds AS NOT MATERIALIZED (
    SELECT ST_TileEnvelope(@zoom, @x, @y) AS tile
),
mvtgeom AS (
    SELECT node_id, ST_AsMVTGeom(geom, tile, {{ $extent }}) AS geom, tags
    FROM osm_nodes, bounds
    WHERE ST_Intersects(geom, tile)
    {{ if .WithKey }}
        AND tags ? @key
    {{ if .WithValue }}
        AND tags->>@key = @value
    {{- end -}}
    {{- else if .WithValue -}}
        AND jsonb_path_exists(tags, ('$.keyvalue()[*] ? (@.value == "'
        || @value || '"')::jsonpath)
    {{- end -}}
)
SELECT ST_AsMVT(mvtgeom, 'nodes', {{ $extent }}, 'geom', 'node_id'), count(*)
FROM mvtgeom

```

Server (2nd try)

much simpler to configure
and
to deploy

Caching tiles?

in database? in file system? nginx?

but: cache invalidation

turns out: **probably fast enough without cache**

5min cache in browser

What I learned about OSM



Object types

N ☒ W ☒ R ☒

Tag filter

Key:

Value:

Filter

Clear

Zoom in for interactive map

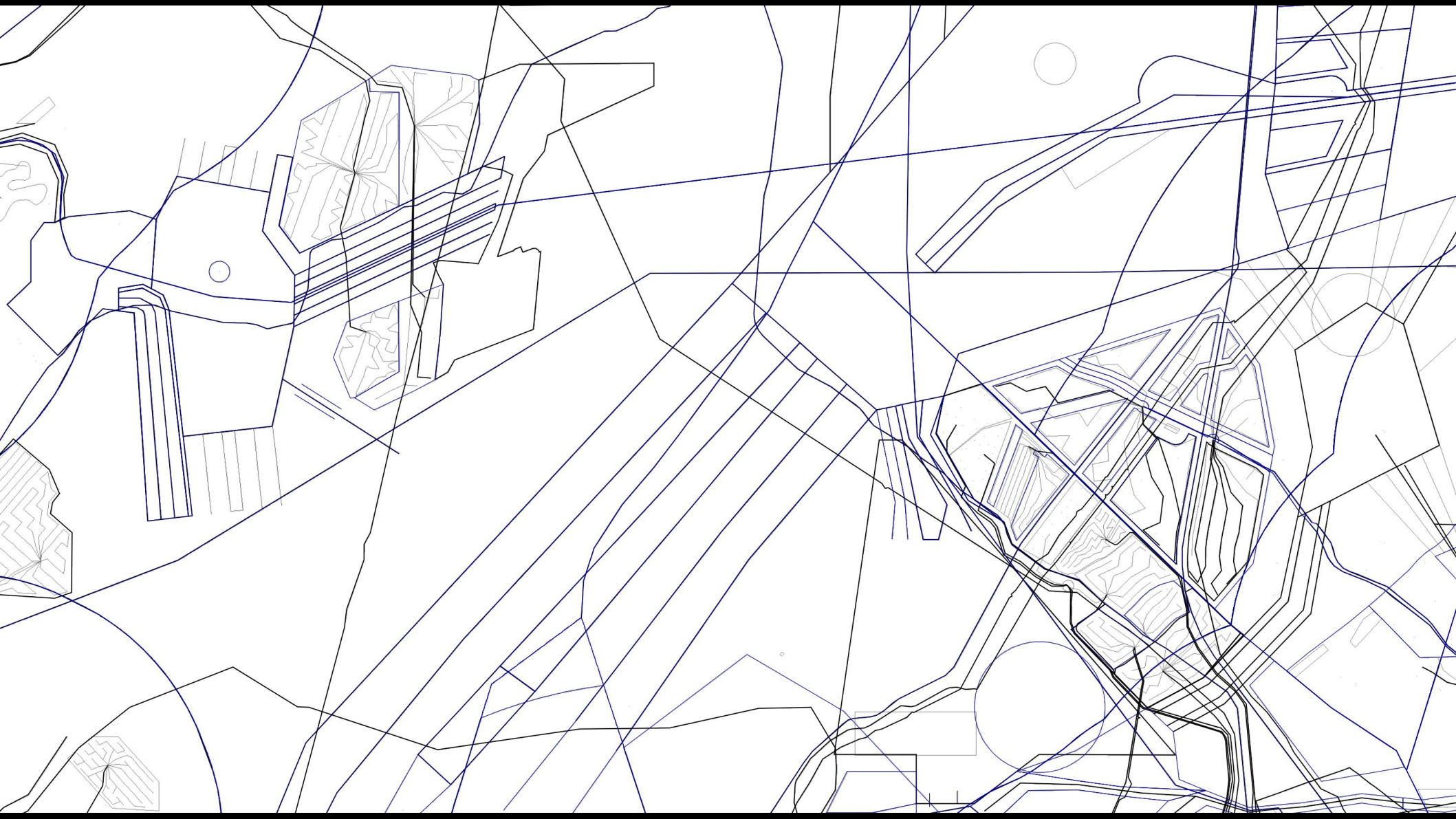
Selected: 1 (zoom in to show selection on map)

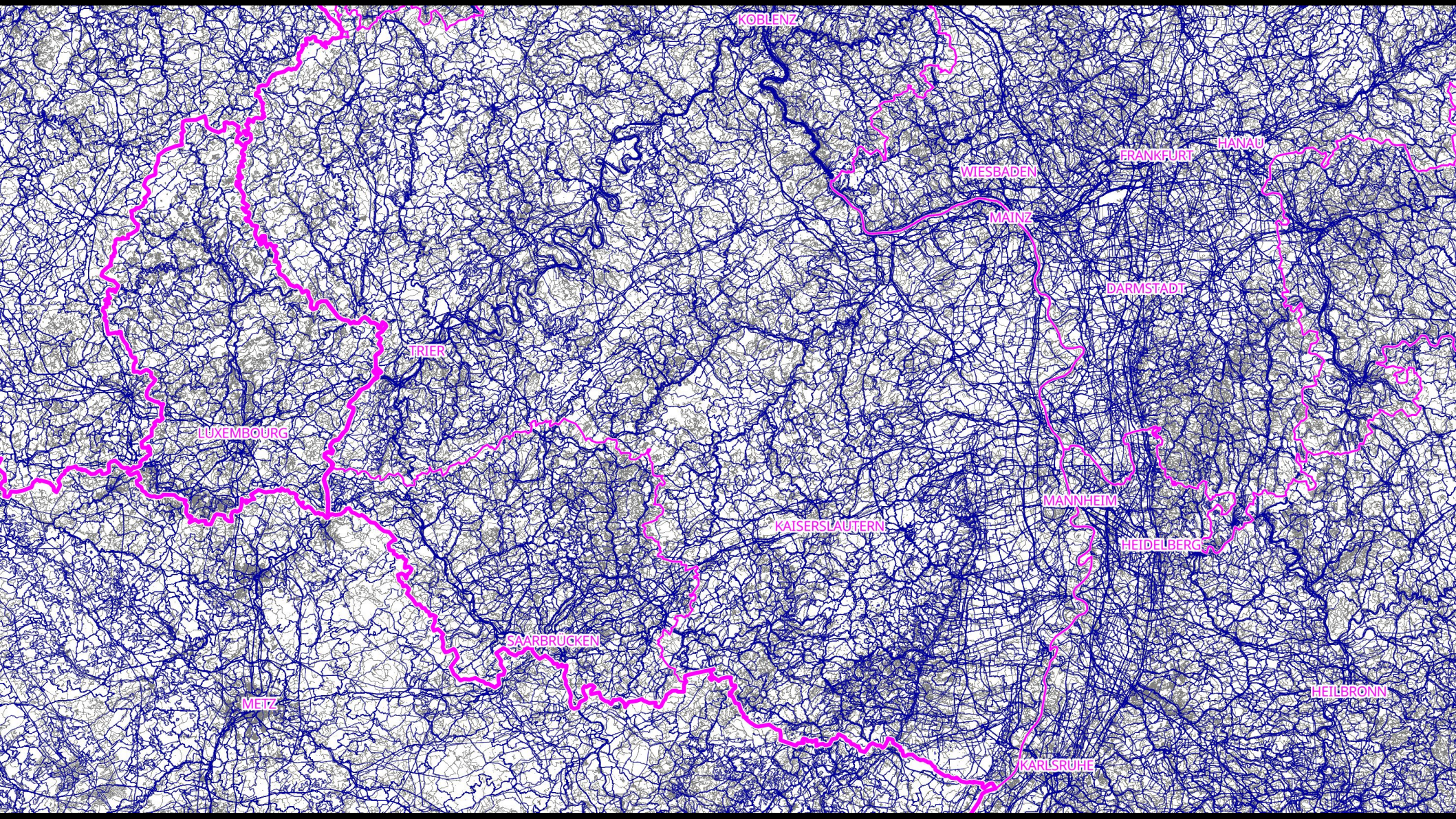
1

way 1094680663

alt_name	Pacific trash vortex
material	plastic
name	Great Pacific eastern garbage patch
short_name	GPGP
start_date	1997
wikidata	Q638904







What's next?

find production server!

filtering in small zoom levels?

color-coded maps?

THANKS FOR YOUR INTEREST!

JOCHEN TOPF – JOCHEN@TOPF.ORG

<http://test.osm2pgsql.org>
<https://codeberg.org/jot/osm-spyglass>